

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : Mechanical Semester : VI Subject Code & Name: BTMEC604B IC Engine Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following. (This is just a sample instruction)		
A)	Differentiate between C.I engine and S.I engine.	(Analyze)	6
B)	Compare the efficiencies of Otto Cycle, Diesel Cycle and Dual Cycle on different aspects with the help of P-V and T-S diagrams.	(Analyze)	6
C)	Give advantages and drawbacks of electric vehicles	(Understand)	6
Q.2	Solve Any Two of the following. (This is just a sample instruction)		
A)	Write a note on ignition delay in CI engine	(Understand)	6
B)	Describe the following terms i) Flash Point ii) Fire Point iii) Pour Point	(Understand)	6
C)	Explain important qualities and ratings of SI Engines fuels	(Analyze)	6
Q. 3	Solve Any One of the following. (This is just a sample instruction)		
A)	Explain the working of a battery ignition system with neat sketch	(Analyze)	6
B)	What are desirable properties of good lubricating oil?	(Understand)	6
C)	Explain engine air cooling system with neat sketch.	(Understand)	6
Q.4	Solve Any Two of the following. (This is just a sample instruction)		
A)	Give types of supercharger and explain any one type with sketch	(Analyze)	6
B)	A two stroke cycle internal combustion engine has a mean effective pressure of 5.8 bar. The speed of the engine is 940 rpm. If the diameter of piston and stroke are 120 mm and 130 mm respectively, find the indicated power developed and piston speed.	(Application)	6

C)	State different exhaust emissions of I.C. engine and explains its effect on human health.	(Understand)	6
Q. 5	Solve Any One of the following. <i>(This is just a sample instruction)</i>		
A)	Explain fuel cell technology in brief with neat sketch	(Understand)	6
B)	Compare hybrid vehicles and electric vehicles	(Understand)	6
C)	State the different alternative fuels and its advantages over conventional fuels.	(Application)	6
	*** End ***		

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